Inequalities – new problems

Problems built on recent research results

Each problem is a special case of a published result or of a lemma used in its proof; the papers are in this folder. The problems are ordered by difficulty. The difficulty is the position of the problem in an IMC day: 1 easy, 2 below average, 3 medium, 4 above average, 5 hard.

Problem 1. *IMC level: 1/5 (easy)*

Let X be a real random variable such that $\mathbb{E}\, e^{tX} \leqslant e^{t^2/2}$ for all $t \in \mathbb{R}$. Prove that

$$\mathbb{E}\, e^{3X^2/8} \leqslant 2,$$

and show that $\frac{3}{8}$ cannot be replaced by any larger constant.

Based on: L. Leskelä, M. Zhukov, *Sharp constants relating the sub-Gaussian norm and the sub-Gaussian parameter*, arXiv:2507.05928, Theorem 1.1.

Problem 2. *IMC level: 2/5 (below average)*

For vectors $x, y, z \in \mathbb{R}^3$ put

$$S = \|x\|^2\langle y,z\rangle^2 + \|y\|^2\langle z,x\rangle^2 + \|z\|^2\langle x,y\rangle^2, \qquad P = \langle x,y\rangle\langle y,z\rangle\langle z,x\rangle.$$

Determine all real c such that $S + cP \geqslant 0$ for all x, y, z.

Based on: N. El Idrissi, H. Zoubeir, *Two Hornich–Hlawka-type and Gram matrix-based inequalities*, arXiv:2601.18768, Section 3, inequalities (2) and (3).

Problem 3. *IMC level: 2/5 (below average)*

(a) Let $\alpha \geqslant 1$ and let $f\colon [0,1] \to [0,\infty)$ be continuous and non-decreasing. Prove that

$$\alpha \int_0^1 (1-x)^{\alpha-1} f(x)^\alpha \, dx \leqslant \Big(\int_0^1 f(x)\, dx\Big)^\alpha.$$

(b) Prove the same inequality for $0 < \alpha \leqslant 1$ and continuous non-increasing $f\colon [0,1] \to [0,\infty)$.

(c) Show that for $\alpha > 1$ equality in (a) holds only for constant f.

Based on: Ł. Płociniczak, *The Bushell–Okrasiński inequality*, arXiv:2203.15256, inequality (20), Theorem 2 and Corollary 1.

Problem 4. *IMC level: 2/5 (below average)*

Let $A, B \subseteq [0,1]$ be finite unions of intervals of total lengths a and b. Prove that

$$\int_A\int_B \cos\big(2\pi(x-y)\big)\, dx\, dy \leqslant \frac{\sin(\pi a)\sin(\pi b)}{\pi^2}.$$

Based on: H. Hajaiej, *On the necessity of the assumptions used to prove Hardy–Littlewood and Riesz rearrangement inequalities*, arXiv:1003.3166, inequalities (1.5)–(1.6).

Problem 5. *IMC level: 2/5 (below average)*

Let $d \geqslant 1$ and let $a_1, \ldots, a_d$ be real numbers; put $a_0 = a_{d+1} = 0$. Prove that

$$\Big(\sum_{j=1}^{d} |a_j|\Big)^2 \leqslant \frac{d(d+1)(d+2)}{12} \sum_{j=1}^{d+1} (a_j - a_{j-1})^2,$$

and find all cases of equality.

Based on: R. Hindov, S. Nitzan, J.-F. Olsen, E. Rydhe, *A sharp discrete Sobolev inequality and left inverses of higher order differences*, arXiv:2610.19010, Theorem 1 ($\alpha = 1$).

Problem 6. *IMC level: 2/5 (below average)*

Let $f\colon [0,1] \to \mathbb{R}$ be continuously differentiable with $m \leqslant f'(x) \leqslant M$ for all x. Prove that for every $t \in [0,1]$

$$\Big| f(t) - \int_0^1 f(x)\,dx - \Big(t - \frac{1}{2}\Big)\big(f(1) - f(0)\big) \Big| \leqslant \frac{M-m}{8},$$

and that for every fixed t the constant $\frac{1}{8}$ cannot be replaced by a smaller one.

Based on: K. Castillo, Â. Macedo, *On a measure-theoretic reading of β-Grüss-type inequalities*, arXiv:2607.06439, Theorem 2.8; cf. D. R. Anderson, arXiv:1409.5888, Corollary 7.3 (constant $\frac{1}{4}$).

Problem 7. *IMC level: 2/5 (below average)*

Let $f\colon [0,1]^2 \to \mathbb{R}$ be continuous and such that

$$f(x_2, y_2) + f(x_1, y_1) \geqslant f(x_1, y_2) + f(x_2, y_1) \qquad \text{whenever } 0 \leqslant x_1 \leqslant x_2 \leqslant 1,\ 0 \leqslant y_1 \leqslant y_2 \leqslant 1.$$

Prove that

$$\int_0^1 f(x, 1-x)\,dx \leqslant \int_0^1 \int_0^1 f(x,y)\,dx\,dy \leqslant \int_0^1 f(x,x)\,dx.$$

Based on: A. Komisarski, T. Rajba, *On Popoviciu's concept of convexity for functions of d variables*, arXiv:2404.13723, Theorem 45 ($d = 2$, $\mathbf{n} = (1,1)$).

Problem 8. *IMC level: 2/5 (below average)*

For a complex $n \times n$ matrix A let $w(A) = \max\{|x^* A x| : x \in \mathbb{C}^n,\ \|x\| = 1\}$ be its numerical radius and let $\|\cdot\|$ be the operator norm. Prove that

$$\frac{1}{4}\|A^*A + AA^*\| \leqslant w(A)^2 \leqslant \frac{1}{2}\|A^*A + AA^*\|,$$

and show that both constants $\frac{1}{4}$ and $\frac{1}{2}$ are attained for suitable 2×2 matrices $A \neq 0$.

Based on: F. P. Najafabadi, H. R. Moradi, *Advanced refinements of numerical radius inequalities*, arXiv:2011.08443, inequalities (1.5) and (1.6).

Problem 9. *IMC level: 2/5 (below average)*

(a) Let $p > 1$ and let X, Y be independent identically distributed random variables with $\mathbb{E}X = 0$ and $\mathbb{E}|X|^p = 1$. Prove that

$$\mathbb{E}\max(X, Y) \leqslant \Big(\frac{p-1}{2p-1}\Big)^{\frac{p-1}{p}}.$$

(b) Deduce that for independent identically distributed X, Y with variance σ^2 one has $\mathbb{E}|X-Y| \leqslant \frac{2}{\sqrt{3}}\,\sigma$, with equality for the uniform distribution on an interval.

Based on: R. Vila, N. Balakrishnan, H. Saulo, *An upper bound and a characterization for Gini's mean difference based on correlated random variables*, arXiv:2301.07229, Section 2, inequality (5).

Problem 10. *IMC level: 2/5 (below average)*

Let $n \geqslant 1$ and let $f\colon \{0,1\}^n \to \mathbb{R}$ be arbitrary. For $x \in \{0,1\}^n$ let $\bar{x}$ be the vector with all coordinates flipped. Prove that

$$\sum_{\{x,\bar{x}\}} \big(f(x) - f(\bar{x})\big)^2 \leqslant \sum_{\{x,y\}} \big(f(x) - f(y)\big)^2,$$

where the left sum runs over the 2^{n-1} antipodal pairs and the right sum over the $n2^{n-1}$ pairs $\{x, y\}$ differing in exactly one coordinate.

Based on: J. Baslingker, B. Virág, *The Efron–Stein inequality for identically distributed pairs*, arXiv:2605.05388, Theorem 2.

Problem 11. *IMC level: 3/5 (medium)*

Find all $p > 0$ such that

$$|a|^p + |b|^p + |c|^p + |a+b+c|^p \;\geqslant\; |a+b|^p + |b+c|^p + |c+a|^p$$

for all real numbers a, b, c.

Based on: D.-Ş. Marinescu, C. P. Niculescu, *A survey of the Hornich–Hlawka inequality*, arXiv:2407.03278, inequality (2.1), Corollary 4 and Section 10.

Problem 12. *IMC level: 3/5 (medium)*

Let $n \geqslant 2$ and let $x_1, \ldots, x_n$ be real numbers with $x_1 + \cdots + x_n = 0$. Prove that

$$n(n-1)\sum_{i=1}^{n} x_i^4 \;\leqslant\; (n^2 - 3n + 3)\Big(\sum_{i=1}^{n} x_i^2\Big)^2,$$

and determine when equality holds.

Based on: J. S. Case, A. Tyrrell, arXiv:2305.03861, Theorem 1.1 (applied to diagonal matrices) and Proposition 2.3.

Problem 13. *IMC level: 3/5 (medium)*

Let $n \geqslant 3$, let $x_1, \ldots, x_n$ be arbitrary real numbers and let $\alpha \in \mathbb{R}$. Put $E_j = e_j(x_1, \ldots, x_n)/\binom{n}{j}$, where e_j is the j-th elementary symmetric polynomial and $E_0 = 1$. Prove that for $1 \leqslant k \leqslant n-2$

$$\big(\alpha E_k + E_{k+1}\big)^2 \;\geqslant\; \big(\alpha E_{k-1} + E_k\big)\big(\alpha E_{k+1} + E_{k+2}\big).$$

Based on: C. Ren, *A generalization of Newton–Maclaurin's inequalities*, arXiv:2205.00873, Theorem 1.2.

Problem 14. *IMC level: 3/5 (medium)*

Let $f, g\colon [0,1] \to [0,\infty)$ be continuous, not both identically zero. Put

$$A = \int_0^1 f^4 + \int_0^1 g^4, \qquad t = \sqrt{\frac{2\int_0^1 f^2 g^2}{A}}.$$

Prove that $\displaystyle\int_0^1 (f+g)^4 \leqslant (1+t)^3 A$.

Based on: E. Carlen, R. Frank, P. Ivanisvili, E. Lieb, *Inequalities for L^p-norms that sharpen the triangle inequality and complement Hanner's inequality*, arXiv:1807.05599, Theorem 1.1 ($p = 4$).

Problem 15. *IMC level: 3/5 (medium)*

Prove that for every positive integer n and every real $x \geqslant 0$

$$\sum_{j=0}^{n-1} \frac{x^{2j}}{(j!)^2} \leqslant \exp\Big(2x\cos\frac{\pi}{n+1}\Big).$$

Based on: B. Bauer, S. Gerhold, *The Fan–Taussky–Todd inequalities and the Lumer–Phillips theorem*, arXiv:2305.14015, inequality (3.3).

Problem 16. *IMC level: 3/5 (medium)*

For a finite simple graph G let $i(G)$ be the number of independent sets of G (the empty set included). The graph $G \times K_2$ has vertex set $V(G) \times \{0,1\}$, and (u,a) is adjacent to (v,b) if and only if $uv \in E(G)$ and $a \neq b$. Prove that

$$i(G)^2 \leqslant i(G \times K_2).$$

Based on: D. Chheda, R. Goel, E. Qiao, *Number of independent sets in regular and irregular graphs: a 31 year journey*, arXiv:2405.18815, Lemma 3.2 and Section 4.

Problem 17. *IMC level: 4/5 (above average)*

Let $f\colon \mathbb{R} \to \mathbb{R}$ be continuous, 1-periodic and piecewise linear (with finitely many linear pieces on $[0,1]$). Let $g\colon \mathbb{R} \to \mathbb{R}$ be the continuous 1-periodic even function, non-increasing on $[0, \frac{1}{2}]$, such that for every $t \in \mathbb{R}$ the sets $\{x \in [0,1) : f(x) > t\}$ and $\{x \in [0,1) : g(x) > t\}$ have the same total length (you may assume that g exists, is unique and is piecewise linear). Prove that

$$\int_0^1 g'(x)^2\,dx \leqslant \int_0^1 f'(x)^2\,dx.$$

Based on: S. Barza, M. Lind, *A variation on the Pólya–Szegő principle in one dimension*, arXiv:2607.03450, inequality (1.2); P. Le, arXiv:2303.02709, Section 5.

Problem 18. *IMC level: 4/5 (above average)*

Let $a_1, \dots, a_m$ be real numbers and for $k \geqslant 0$ let

$$M_{2k} = \frac{1}{2^m} \sum_{\varepsilon \in \{-1,1\}^m} \Big(\sum_{j=1}^m \varepsilon_j a_j\Big)^{2k}.$$

(a) Prove that $M_{2k-2}\, M_{2k+2} \leqslant \dfrac{2k+1}{2k-1}\, M_{2k}^2$ for every $k \geqslant 1$.

(b) Deduce that $M_{2k} \leqslant (2k-1)!!\, M_2^k$ for every $k \geqslant 1$.

Based on: A. Havrilla, P. Nayar, T. Tkocz, *Khinchin-type inequalities via Hadamard's factorisation*, arXiv:2102.09500, Theorem 1 and Section 2.1.

Problem 19. *IMC level: 4/5 (above average)*

Let $v\colon \mathbb{R} \to (0,\infty)$ be continuously differentiable and 2π-periodic. Prove that

$$\Big(\int_0^{2\pi} v^2\,d\theta - 4\int_0^{2\pi} v'^2\,d\theta\Big)\int_0^{2\pi} \frac{d\theta}{v^2} \leqslant 4\pi^2,$$

and that equality holds if and only if $v(\theta)^2 = a + b\cos\theta + c\sin\theta$ for some reals $a > \sqrt{b^2+c^2}$.

Based on: P. Le, *A Lorentz invariant sharp Sobolev inequality on the circle*, arXiv:2303.02709, Theorem 7.1.

Problem 20. *IMC level: 4/5 (above average)*

For real t let $\|t\|$ be the distance from t to the nearest integer and $e(t) = e^{2\pi i t}$. Let $N \geqslant 1$, let $z_1, \dots, z_N$ be complex numbers, let $0 < \delta \leqslant 1$, and let $x_1, \dots, x_R$ be real numbers with $\|x_r - x_s\| \geqslant \delta$ for all $r \neq s$. Prove that

$$\sum_{r=1}^R \Big|\sum_{n=1}^N z_n\, e(n x_r)\Big|^2 \leqslant \Big(\frac{1}{\delta} + \pi(N-1)\Big)\sum_{n=1}^N |z_n|^2.$$

Based on: S. Baier, S. B. Lynch, *Large sieve inequalities with power moduli and Waring's problem*, arXiv:2303.07613, inequality (1) (in a weaker form).

Problem 21. *IMC level: 4/5 (above average)*

Let $h(x) = -x\ln x - (1-x)\ln(1-x)$ for $0 < x < 1$ and $h(0) = h(1) = 0$, and let $\varphi = \frac{1+\sqrt{5}}{2}$. Prove that

$$h(x^2) \geqslant \varphi\, x\, h(x) \qquad \text{for all } x \in [0,1],$$

with equality only for $x \in \{0, 1/\varphi, 1\}$.

Based on: B. S. Ho, *A generalization of Boppana's entropy inequality*, arXiv:2601.19327, Proposition 1 and Theorem 1 ($k = 2$).

Problem 22. *IMC level: 4/5 (above average)*

Let X and Y be independent non-negative random variables, each taking finitely many values, with $\mathbb{E}X \leqslant 1$ and $\mathbb{E}Y \leqslant 1$. Prove that

$$\mathbb{P}(X + Y < 3) \geqslant \frac{4}{9},$$

and that the bound is attained.

Based on: A. Milojević, B. Sudakov, *Perfect matchings in hypergraphs and Feige's inequality*, arXiv:2610.10380, Theorem 2 (the case $n = 2$ of Feige's inequality).

Problem 23. *IMC level: 5/5 (hard)*

Let $z_1, \dots, z_n$ be nonzero complex numbers, let $P(z) = z\prod_{j=1}^{n}(z - z_j)$, and let $w_1, \dots, w_n$ be the zeros of P', counted with multiplicity. Prove that all $w_j \neq 0$ and that

$$\sum_{j=1}^{n} \frac{1}{|w_j|^2} \;\leqslant\; 3\sum_{j=1}^{n} \frac{1}{|z_j|^2} + \Big|\sum_{j=1}^{n} \frac{1}{z_j}\Big|^2.$$

Based on: O. Holtz, *A reciprocal Schoenberg inequality and its extensions*, arXiv:2610.41325, inequality (1.3).

Problem 24. *IMC level: 5/5 (hard)*

Let $f\colon [0,1] \to \mathbb{R}$ be continuous and assume that for every interval $[a,b] \subseteq [0,1]$ with $a < b$

$$\frac{1}{b-a}\int_a^b f(x)\,dx - \min_{[a,b]} f \leqslant 1.$$

(a) Prove that for every $\lambda \geqslant 1$ the set $\{x \in [0,1] : f(x) - \min_{[0,1]} f > \lambda\}$ has total length at most $e^{1-\lambda}$.

(b) Show that the rate is optimal: there are no constants $c > 1$ and C such that this length is at most $Ce^{-c\lambda}$ for all such f and all $\lambda \geqslant 1$.

Based on: C. Zhang, *Sharp constants in the one-sided John–Nirenberg inequality for functions of bounded lower oscillation*, arXiv:2608.10892, Theorem 1.1.

Problem 25. *IMC level: 5/5 (hard)*

Let A and B be complex $n \times n$ matrices with operator norms $\|A\| < 1$ and $\|B\| < 1$. Prove that

$$\det(I - A^*A)\,\det(I - B^*B) + |\det(A - B)|^2 \leqslant |\det(I - A^*B)|^2.$$

Based on: B.-Y. Xi, F. Zhang, *Some matrix inequalities of log-majorization type*, arXiv:2103.06174, Theorem 4.1, inequality (4.1).